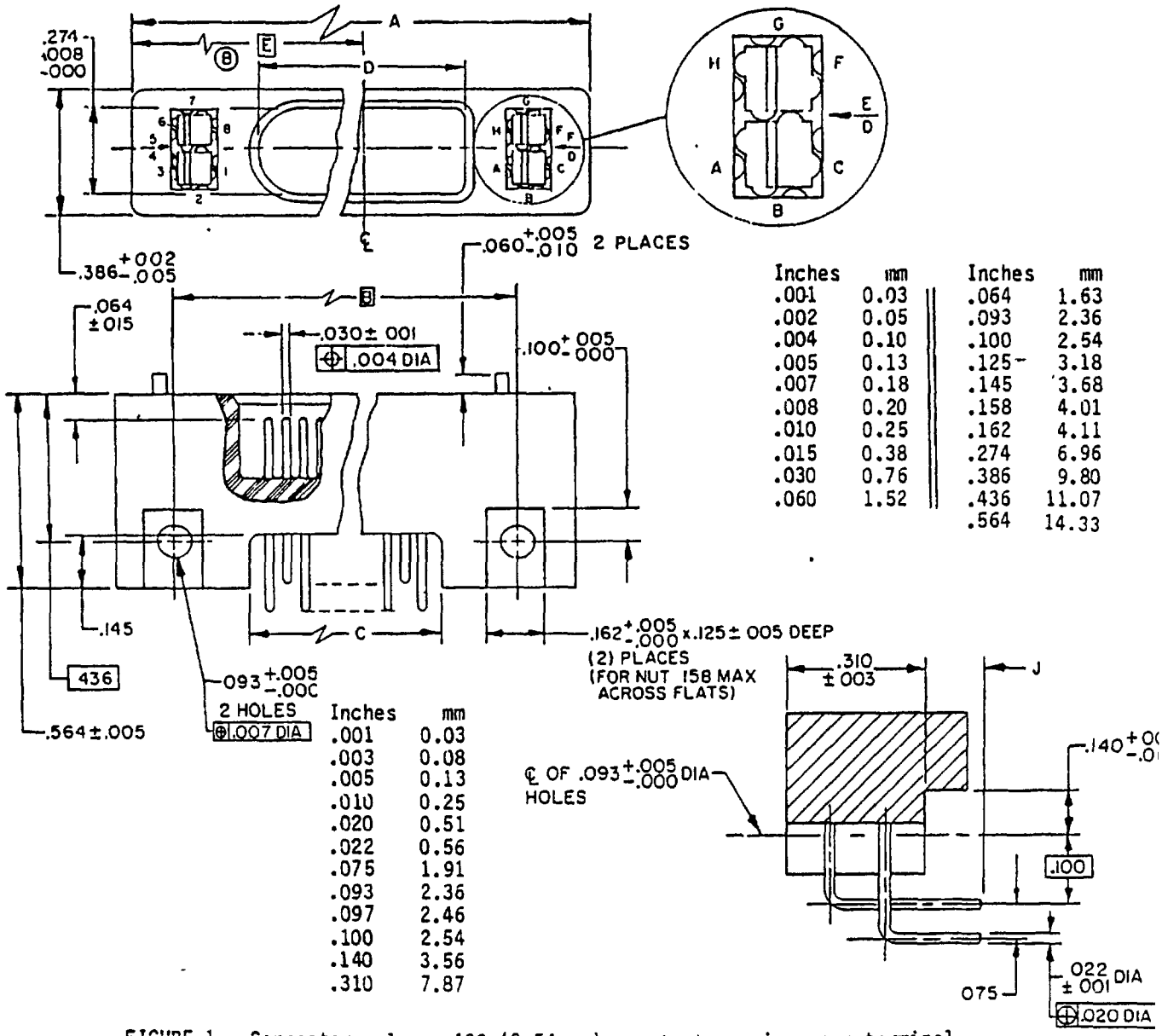
MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, PIN, RIGHT ANGLE, PLUG, ELECTRICAL, POLARIZED FOR PRINTED WIRING BOARDS (.100 SPACING)

Ⓒ This specification inactive for new design after 1 October 1986

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connector described herein shall consist of this specification and the latest issue of MIL-C-55302.



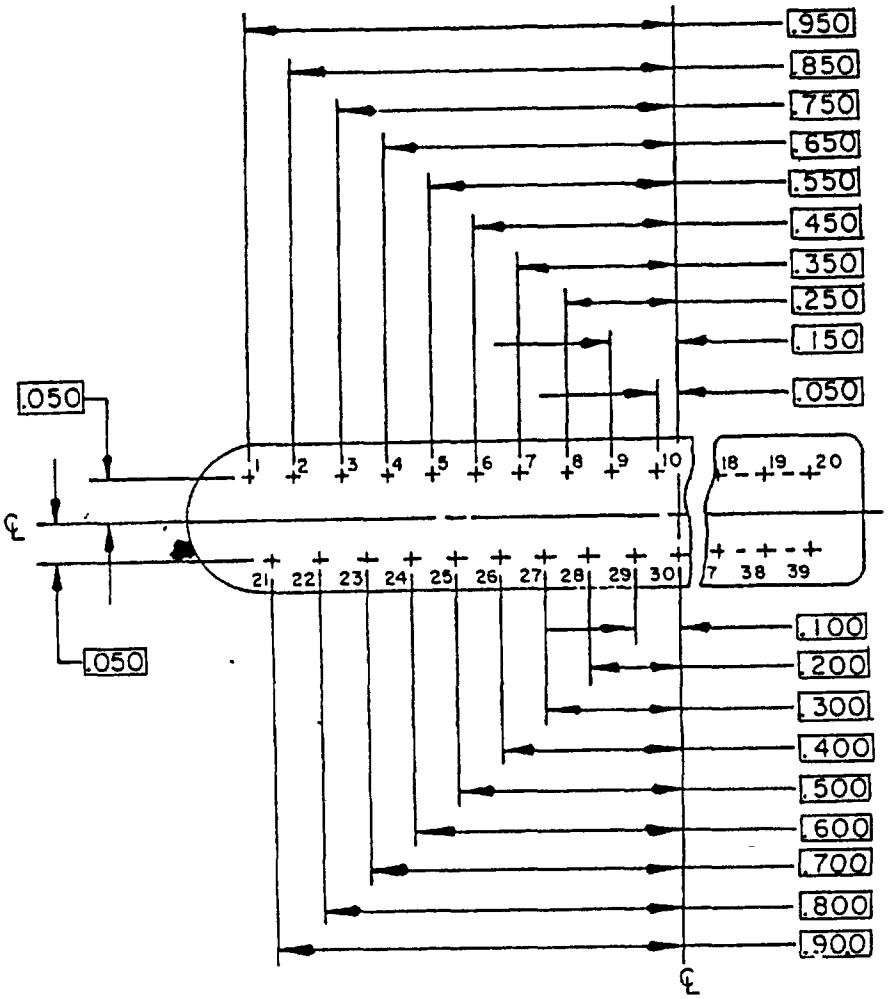
Inches	mm	Inches	mm
.001	0.03	.064	1.63
.002	0.05	.093	2.36
.004	0.10	.100	2.54
.005	0.13	.125	3.18
.007	0.18	.145	3.68
.008	0.20	.158	4.01
.010	0.25	.162	4.11
.015	0.38	.274	6.96
.030	0.76	.386	9.80
.060	1.52	.436	11.07
		.564	14.33

Inches	mm
.001	0.03
.003	0.08
.005	0.13
.010	0.25
.020	0.51
.022	0.56
.075	1.91
.093	2.36
.097	2.46
.100	2.54
.140	3.56
.310	7.87

FIGURE 1. Connector, plug, .100 (2.54 mm), contact spacing, pin terminal

Ⓒ denotes changes

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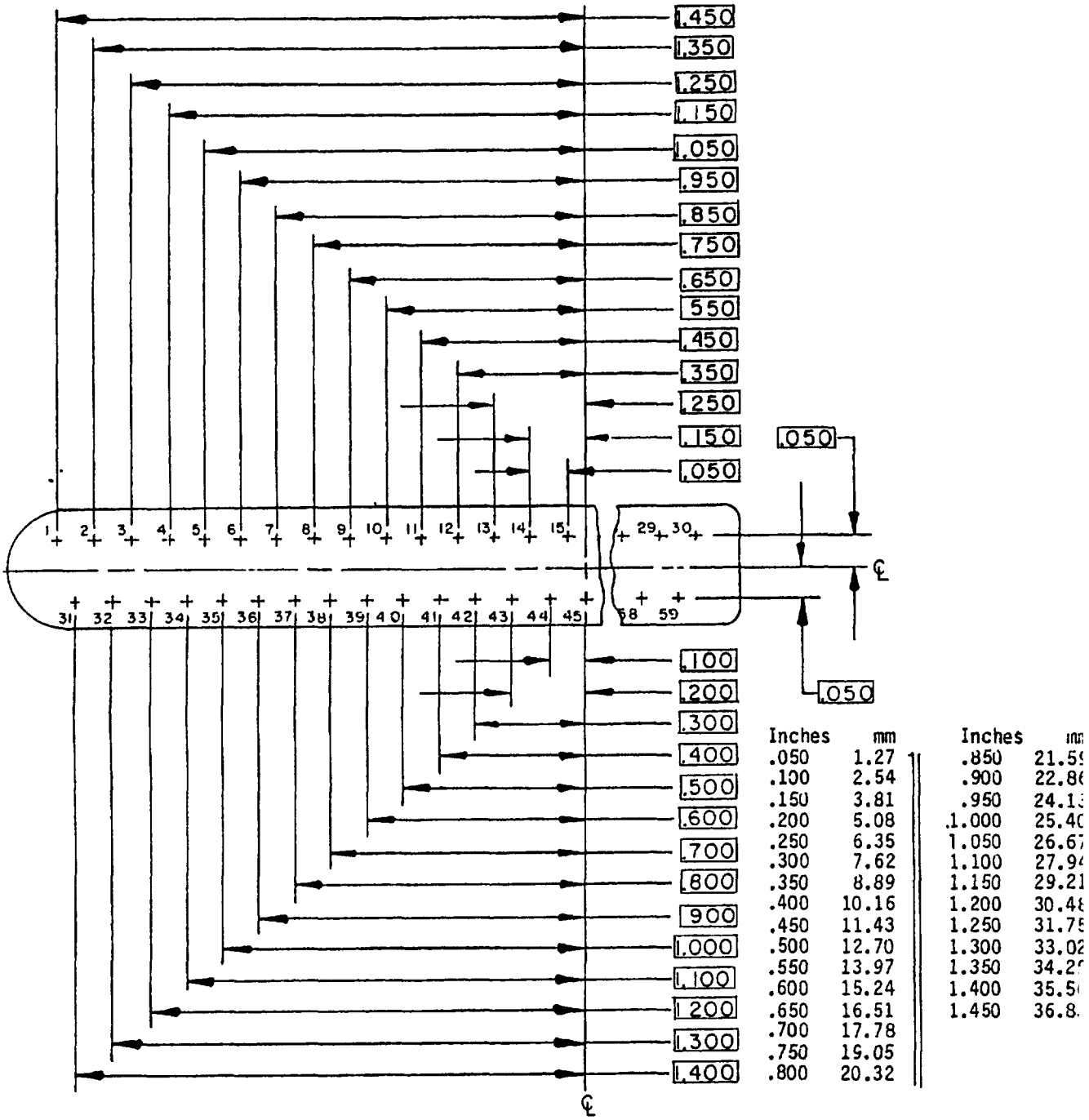
Inches	mm
.050	1.27
.100	2.54
.150	3.81
.200	5.08
.250	6.35
.300	7.62
.350	8.89
.400	10.16
.450	11.43
.500	12.70
.550	13.97
.600	15.24
.650	16.51
.700	17.78
.750	19.05
.800	20.32
.850	21.59
.900	22.86
.950	24.13

NOTE:
Contacts 1, 20, 21, and 39 are identified on the front and rear of the connector per the insert arrangement.

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Nominal spacing between any two adjacent pin contacts shall be .100.
4. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
5. For dimensions A through D see table I.
6. Gold plating shall not be cracked or scratched.
7. Mating connectors provided a total of 256 different polarization positions. Polariation components conforming to MIL-C-55302/78-02 are supplied with the connector uninstalled.

FIGURE 2. Insert arrangement for .100 (2.54 mm) contact spacing connectors. (Male engaging face).

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Inches	mm	Inches	mm
.050	1.27	.850	21.59
.100	2.54	.900	22.86
.150	3.81	.950	24.13
.200	5.08	1.000	25.40
.250	6.35	1.050	26.67
.300	7.62	1.100	27.94
.350	8.89	1.150	29.21
.400	10.16	1.200	30.48
.450	11.43	1.250	31.75
.500	12.70	1.300	33.02
.550	13.97	1.350	34.29
.600	15.24	1.400	35.56
.650	16.51	1.450	36.83
.700	17.78		
.750	19.05		
.800	20.32		

NOTE:
Contacts 1, 30, 31, and 59 are identified on the front and rear face of the connector per the insert arrangements.

FIGURE 2. Insert arrangement for .100 (2.54 mm) contact spacing connectors (Male engaging face) - Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Material: In accordance with MIL-C-55302.

Plating: The contact plating shall be gold in accordance with MIL-G-45204, type II, grade C, class 1 over copper in accordance with MIL-C-14550, class 4.

Contact identification: Contact locations are identified numerically as shown on figure 2.

Mating and unmating: The maximum mating force, in pounds, shall be 0.5 times the number of contacts, and the withdrawal force, in pounds, shall be a minimum of 0.08 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: With a test current of 3 A dc, the average resistance of all contact pairs measured shall not exceed 0.010 ohm and no individual contact pair shall have a resistance exceeding 0.020 ohm.

Dielectric withstanding voltage:

Sea level: 1,300 V rms, 60 Hz, ac.

High altitude: 325 V rms, 60 Hz, ac.

Contact current rating: 3.0 amperes.

Insert arrangement: See figure 2.

High potential test voltage: 1,300 V rms, at sea level.

Mating connectors: See MIL-C-55302/75.

Part number: M55302/74-(dash number from table I).

TABLE I. Dash number and dimensions.

Dash number	Dimensions 1/						No. of contacts
	A ±.005 (.013)	B	C Min.	D +.008(.20) -.000	E	J +.035(.89) -.020(.51)	
01	2.700 (68.58)	2.400 (60.96)	2.040 (51.82)	2.093 (53.16)	1.200 (30.48)	.097 (2.46)	39
02	3.700 (93.98)	3.400 (86.36)	3.040 (77.22)	3.093 (78.56)	1.700 (43.18)	.097 (2.46)	59
03	2.700 (68.58)	2.400 (60.96)	2.040 (51.82)	2.093 (53.16)	1.200 (30.48)	.159 (4.04)	39
04	3.700 (93.98)	3.400 (86.36)	3.040 (77.22)	3.093 (78.56)	1.700 (43.18)	.159 (4.04)	59

1/ Metric equivalent are given for general information only. Millimeters are in parentheses.

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Custodian:
Air Force - 17

Review activities:
Air Force - 11, 17, 85
DLA - ES

User activity:
Air Force - 19

Preparing activity
Air Force - 17

Agent:
DLA - ES

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